What happens to the CO_2 once injected underground?

The CO_2 is held in the small pore spaces present in rocks. The ideal saline reservoir for geological storage shares characteristics with the best oil producing rocks. This is sedimentary rock with good porosity (percentage of open pore spaces) and permeability (connectivity of open pore space).

Directly above the storage reservoir, at depths of at least 5,500 ft, is a relatively thick caprock unit (hundreds of feet thick) of very low porosity and permeability, such as shale. The caprock layer acts as a lid or a cover for the saline reservoir. The depth of the reservoir is also important. A suitable saline reservoir will be thousands of feet below drinking water levels yet economically viable for drilling.

Also, the volume of the potential storage space must be great enough to justify the expense of drilling a well, and having more than one shale caprock which will help ensure containment of the CO_2. The Illinois Basin - Decatur Site will store CO_2 more than a mile below the surface, and have two backup shale seals in addition to the primary shale seal in the mile between the Mt. Simon Sandstone and the surface.







What happens to the CO_2 in the event of an earthquake?

Earthquake energy traveling through bedrock is many times smaller than what we experience on the ground surface. In the event of an earthquake, the CO_2 in the porous spaces of the rock may react slightly to the space being momentarily compressed. But, because the Mt. Simon has good porosity and permeability, fluids can easily move laterally without damaging reservoir containment.

In water wells, a very small, momentary change in water levels because of earthquake waves passing through the area of the well has occasionally been recorded. This is believed to be caused by the earthquake wave slightly compressing the rock as it passes through, and then letting the bedrock relax back to normal.